Examples : Phenacite - Be_2SiO_4 (Be^{2+} ions are tetrahedrally surrounded by O^{2-} ions), Olivine - $(Fe/Mg)_2SiO_4$ (Fe^{2+} and Mg^{2+} cations are octahedrally surrounded by O^{2-} ions),

Pyro silicate (or) Soro silicates): Silicates which contain $[Si_2O_7]^{6-}$ ions are called pyro silicates (or) Soro silicates. They are formed by joining two $[SiO_4]^{4-}$ tetrahedral units by sharing one oxygen atom at one corner.(one oxygen is removed while joining). Example : Thortveitite - $Sc_2Si_2O_7$

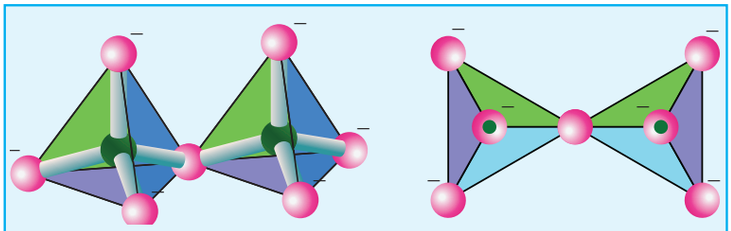


Figure 2.12 Structure of Pyro silicate

Cyclic silicates (or Ring silicates)

Silicates which contain $(SiO_3)_n^{2n-}$ ions which are formed by linking three or more tetrahedral SiO_4^{4-} units cyclically are called cyclic silicates. Each silicate unit shares two of its oxygen atoms with other units.

Example: Beryl $[Be_3Al_2(SiO_3)_6]$ (an aluminosilicate with each aluminium is surrounded by 6 oxygen atoms octahedrally)

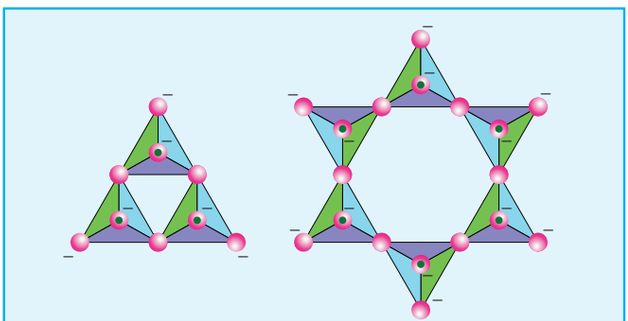


Figure 2.13 Structure of Cyclic silicates

Inosilicates :

Silicates which contain 'n' number of silicate units linked by sharing two or more oxygen atoms are called inosilicates. They are further classified as chain silicates and double chain silicates.

Chain silicates (or pyroxenes): These silicates contain $[(SiO_3)_n]^{2n-}$ ions formed by linking 'n' number of tetrahedral $[SiO_4]^{4-}$ units linearly. Each silicate unit shares two of its oxygen atoms with other units.

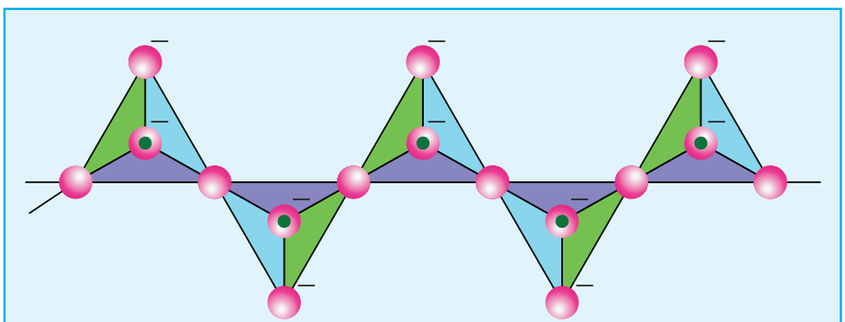


Figure 2.14 Structure of Chain silicates

Example: Spodumene - $LiAl(SiO_3)_2$.

Double chain silicates (or amphiboles): These silicates contains $[Si_4O_{11}]_n^{6n-}$ ions. In these silicates there are two different types of tetrahedra : (i) Those sharing 3 vertices (ii) those sharing only 2 vertices.

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